

Research article

Local banks and flood risk: The case of Germany

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ABSTRACT

This paper uses a simulation model to evaluate the effects of river flooding events occurring within Germany on *regional banks*. Under a 1.5 °C increase in temperature, the impact is overall rather small, even accounting for the devaluation of loans exposed to floods. Specifically, climate related risks would increase bank losses by 0.1% with large variability across NUTS2 regions. However, under a 3 °C increase, bank losses increase up to 1% of total assets. We show that the implementation of adaptation solutions would be successful in keeping risks at the current level. By supporting adaptation initiatives, such as investing in climate-resilient infrastructure or providing financial services to support sustainable practices, banks can play a crucial role in building the resilience of the financial sector and reducing the overall impact on the economy.

1. Introduction

The climate crisis has emerged as a significant threat to firms around the world, impacting not only their operations but also their debt repayment capacity. As extreme weather events become more frequent and severe, companies face an increased risk of being unable to meet their financial obligations, thus increasing their probability of default. These disruptive events carry negative implications for their lending institutions. When a business faces economic repercussions, their capacity to pay bank their loans is compromised, which in turn jeopardizes the credit quality of banks' loan portfolio leading to higher default rates and loan losses. Additionally, banks could face possible asset devaluations, due to properties located in areas prone to climate-related disasters, such as floods or hurricanes, which could suffer a decline in value due to increased risk perceptions. Understanding and managing climate-related financial risks arising from the escalating frequency and intensity of catastrophic events, called physical risk, is thus crucial for maintaining financial stability. It requires assessing the vulnerabilities of financial institutions, integrating climate-related factors into risk management frameworks, and promoting climate resilience and adaptation measures. The nascent literature shows some mixed evidence on the impact of weather-related natural disasters on the financial sector. The study by [Blickle et al. \(2021\)](#) finds that large banks are barely affected by extreme weather-induced events and their income even increases due to subsequently higher loan provision. The impact is more pronounced in local banks but still not large enough to cause solvency concerns. The

analysis from [Calice and Miguel Liriano \(2021\)](#) also finds a small, yet statistically significant result of large-scale natural disasters on banks' credit portfolios. In particular, in Latin American and Caribbean countries banks' non-financial corporations' non-performing loans increase by up to 2.5 percentage points post-disaster.

Flood risks present significant concerns and a number of studies have delved into this issue. A study by the Bank of Japan ([Yamamoto and Naka, 2021](#)) identifies a negative impact of flood damages on company profitability (proxied by the ratio of profit to sales) especially in the manufacturing industry. Interestingly, the impact tends to be larger for firms located in places that do not frequently experience floods. Furthermore, flood events can have far-reaching, indirect impacts on companies' financial performance through the global supply chain network. [Haraguchi and Lall \(2015\)](#) document declines in profitability of up to about 60 % in some Japanese automakers due to a flood, which occurred in Thailand. Evidently, such substantial impacts could negatively affect firms' debt repayment capacity with far-reaching repercussions on lending institutions. More related to our work, [Caloia and Jansen \(2021\)](#) find that the Dutch banking sector is sufficiently capitalised to withstand floods in areas where the real estate exposure is low. However, the impact would be larger in case more severe floods hit the densely populated areas of the country. Two studies focus on small, regional financial institutions and empirically examine the indirect impact of floods on them. They both document the adverse effects of floods on banks either by temporarily increasing their impaired loans ([Shala and Schumaker, 2024](#)) or by hindering their profitability

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(Pagliari, 2023).

As a country with a considerable number of rivers, Germany is prone to flood events. There have been significant floods in the past that have affected various regions of the country especially during periods of heavy rainfall as it happened in 2021.¹ A recent study by Shala and Schumaker (2024) shows that the profitability of banks in affected regions was impacted in the years following the flood. In addition, the paper explores how regionally less diversified banks in Germany adjusted their loan loss provisioning following the severe summer flood of 2013, which affected widespread regions mostly in Eastern Germany. The authors highlight the need for taking a forward-looking perspective when accounting for the impact of climate-induced physical risk events on banks' balance sheets.

Policymakers have taken several policy initiatives, which require banks to assess and disclose risks arising from climate-related natural disasters, integrate these risks into their risk management frameworks, and align their investment portfolios and financial products with sustainability criteria. The specific requirements vary for each initiative, but the overall goal is to foster a more climate-resilient financial sector by considering and addressing climate physical risks. At the European Union (EU) level, several policy actions require banks to assess, report and integrate climate-related risks into their risk management frameworks. The Sustainable Finance Strategy² expects banks to conduct comprehensive risk assessments that consider the potential impacts of climate change on their business activities. As part of the Strategy, European supervisory authorities will work on climate stress testing of the financial system including a one-off Fit-for-55 climate risk scenario analysis in cooperation with the European Central Bank (ECB) and the European Systemic Risk Board. The latter aims to assess the resilience of the financial sector and gain insights into its capacity to support the transition to a lower carbon economy even under conditions of stress. The Non-Financial Reporting Directive³ requires banks under its scope to disclose information on their policies, risks, and outcomes related to environmental matters, including climate change and physical risks. The EU Taxonomy,⁴ which provides criteria for determining, which economic activities can be classified as environmentally sustainable, details how banks need to identify and report on investments or products that contribute to climate change adaptation or mitigate physical risks.

The paper contributes to this debate by assessing the effect of localized climate-related events on regional banks in Germany, across various global warming scenarios. In doing so, it aims to shed light on potential vulnerabilities and risks within the German banking sector, paving the way for more informed decision-making, in particular on risk mitigation strategies. To the best of our knowledge, this is the first study to use bank balance sheet data to quantify the size of bank losses triggered by climate related events under different climate warming scenario. In addition, the paper introduces a novel discussion on the effectiveness of adaptation strategies in mitigating financial risks, offering insights for policymakers and financial institutions seeking to

maintain stability in the face of climate change. This evidence provides a valuable contribution to discussions on how the financial sector can actively respond to and prepare for climate-related challenges.

The study focuses on 677 cooperative banks, which are geographically more concentrated in their exposures, given their strong connection to the local economy compared to other types of banks. We assume that the lending activities of a cooperative bank are conducted at the NUTS2 level, which represents relatively large regions in terms of area. Based on 2023 ECB analytical indicators on physical risks,⁵ the size of German bank assets currently at risk due to river floods is 0.6% of their portfolios to non-financial corporations. As the exposure to floods varies within the country, we distribute the amount of assets across different NUTS2 regions, using the share of people exposed as obtained from the Risk Data Hub. To account for the strain on banks' loan portfolios that can arise from the impact of river flood events, we consider the assets at risk as non-performing loans. As global temperatures rise due to global warming, we assume that non-performing loans might increase in proportion to the economic losses caused by floods, as projected in PESETA IV (Feyen et al., 2022; Dottori et al., 2020).

It is important to address the fact that there is large amount of uncertainty associated with the estimates of losses arising from climate-related events. Even when the impact of floods on the business activities is evaluated in absence of future climate changes, many input parameters' variability need to be assessed (see e.g. Alfieri et al., 2018 for uncertainty associated to the spatial resolution of the input data) through sensitivity analysis (e.g. Morris method as in Cappato et al., 2022). This uncertainty becomes even more of an issue, when climate change scenarios are considered. In our case, the analysis for potential future losses is based on the PESETA IV report (Feyen et al., 2022; Dottori et al., 2020), where an ensemble of 22 model realisations characterises the uncertainty regarding future climate projections. However, actual uncertainty might still be under estimated by using the ensemble (see McSweeney et al., 2015).

The present study focuses on two climate change scenarios: a temperature increase up to 1.5 °C, and a more severe one associated to temperature increase up to 3 °C. The former aims at depicting a near-present situation as the latest data on global warming signal that we are reaching the 1.5 °C temperature increase faster than expected, while the latter is used to assess the long-term development with or without adaptation measures.

A micro-simulation portfolio model based on individual bank balance sheet data is used to derive the loss distribution for the banking sector, considering a situation in which losses from physical risk come in addition to initial financial sector losses. Two cases are considered: (a) a normal economic situation where the impact on banks is only due to the physical risk, the materialization of which triggers a small devaluation of assets exposed modelled via a dynamic balance sheet; (b) a crisis type situation where the impact of physical risk is on top of existing financial/economic crisis. For a stable economic situation, the specific impact is rather small and eventually becomes significant under an increase in global average temperature, with losses of around 0.2% of total assets under a 1.5 °C increase in temperature, and around 1% under a 3 °C increase.

When climate risk materializes on top of an existing financial crisis, the magnitude of losses ranges from 0.4% to 0.9% of total assets, depending on the specific climate warming scenarios, and climate related financial risks are concentrated in few regions particularly exposed to climate-related events (up to 5% of total assets). Finally, we investigate the impact of physical risks when adaptation strategies are implemented. The level of risk associated with a 3 °C increase in global warming can be mitigated to the extent that it becomes comparable to the current situation both in crisis time and in normal times. The results

¹ <https://www.bbc.com/news/world-europe-57867773>.

² Communication from the Commission to the European Parliament, the Council, the European Economic and Social Committee and the Committee of the Regions Strategy for Financing the Transition to a Sustainable Economy, COM/2021/390 final.

³ Directive 2014/95/EU of the European Parliament and of the Council of 22 October 2014 amending Directive 2013/34/EU as regards disclosure of non-financial and diversity information by certain large undertakings and groups. OJ L 330, 15.11.2014, p. 1-9.

⁴ Commission Delegated Regulation (EU) 2021/2178 of 6 July 2021 supplementing Regulation (EU) 2020/852 of the European Parliament and of the Council by specifying the content and presentation of information to be disclosed by undertakings subject to Articles 19a or 29a of Directive (2013)/34/EU concerning environmentally sustainable economic activities, and specifying the methodology to comply with that disclosure obligation (Text with EEA relevance) m C/2021/4987 OJ L 443, 10.12.2021, p. 9-67.

⁵ https://www.ecb.europa.eu/pub/pdf/other/ecb.climate_change_indicators202301~47c4bbbc92.en.pdf.

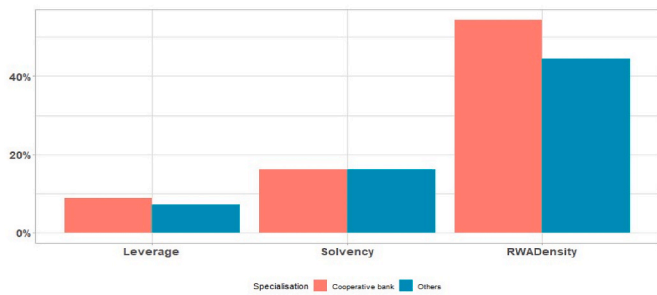


Fig. 1. Risk indicators. The figure displays the comparison of selected risk indicators - Leverage ratio, solvency ratio and the Risk-weighted asset density - for a sample of 677 cooperative banks, and 47 commercial and 356 savings banks (“others”). The source of data is Orbis BankFocus, referring to the year 2022.

indicate that banks operating in regions prone to river floods should take measures to safeguard themselves. One potential approach is to implement additional bank-specific capital add-ons, which would enhance the resilience of the banking sector against these natural-related shocks. This strategy would help mitigate the potential negative impacts on banks’ financial stability and ensure their ability to withstand and recover from such events. Alternatively, banks can contribute to the adaptation efforts aimed at reducing the vulnerability of the regions they operate in. By supporting adaptation initiatives, such as investing in climate-resilient infrastructure or providing financial services to support sustainable practices, banks can play a crucial role, not just in building the resilience of the financial sector, but also in reducing the overall impact on the economy.

The paper is organized as follows. The next section explains how the inputs to the model are derived, notably the share of river floods-related non-performing assets. Section 3 provides an overview of the main modelling framework. Section 4 models a crisis triggered by physical risk, while Section 5 investigates the case of a systemic banking crisis not triggered by physical risk. Finally, Section 6 discusses the role of adaptation strategies in mitigating the risk for the financial sector and Section 7 concludes.

2. Data inputs

The analysis relies on individual balance sheet data for German banks retrieved from Orbis BankFocus as of end of 2020. Specifically, we use Total Assets (TA), Risk Weighted Assets (RWA) and Total Regulatory Capital (K).⁶ By acknowledging their role in financing local economic activities, the study focuses on cooperative banks only. Hence, the final data set covers 677 entities, accounting for around 40% of the total assets in the German banking system. Being inherently more interconnected with local businesses and communities, cooperative banks demonstrate a higher degree of geographical concentration in their exposures and a strong connection to the local economy. This means that their lending activities and investments are focused on specific regions or localities, resulting in a more localized footprint compared to other types of banks.⁷

By comparing our sample with German commercial banks and saving banks, it becomes evident that cooperative banks differ from them in terms of financial riskiness and performance. Notably, cooperative banks exhibit higher levels of risk compared to other types of banks (see Fig. 1), as indicated by lower solvency, and higher RWA density.⁸

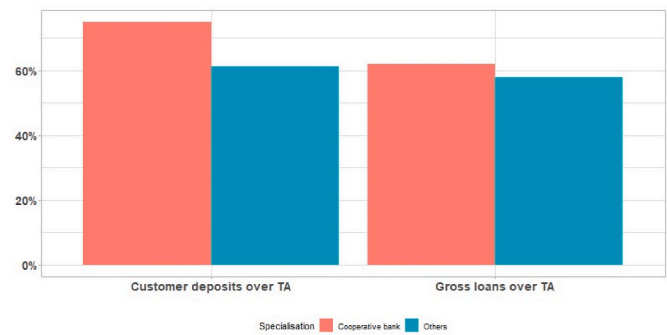


Fig. 2. Business model indicators. The figure displays the comparison of selected business models indicators - customer deposits over total assets, and gross loans over total assets-for a sample of 677 cooperative banks, and 47 commercial and 356 savings banks (“others”). The source of data is Orbis BankFocus, referring to the year 2022.

However, this higher risk is counterbalanced by a stronger capitalization with respect to total assets (leverage), suggesting that cooperative banks have a robust capital base to mitigate potential risks, but probably not sufficient with respect to RWAs (compare leverage and solvency in Fig. 1). One of the key factors contributing to the unique risk profile of cooperative banks is their closer ties to the local economy and their focus on specific sectors or industries. This can expose them to risks associated with local economic downturns and concentration risks in their loan portfolios. Additionally, cooperative banks may face challenges in diversifying their operations beyond their local markets, which can further impact their risk exposure. As a matter of fact, the business model of cooperative banks differs from that of other banks (see Fig. 2). Cooperative banks tend to have a stronger focus on lending activities and rely more heavily on customer deposits. This emphasis on lending and customer relationships aligns with their mission to support local economic activities. In terms of profitability (see Fig. 3), our sample of cooperative banks outperforms commercial and savings banks, as evidenced by higher Return on Assets (and on Equity) and larger operating income.

Together with balance sheet data, a key input to the model is the actual size of assets in banks’ balance sheets that are exposed to non-financial corporations located in river flood-prone areas. The ECB assumes that 0.6% of German financial institutions’ portfolios are at risk considering the intensity of the hazard, while 29.8% are exposed.⁹ Leveraging the ECB’s breakdowns of domestic and cross-border intra euro-area positions, we determine the proportion of assets held by non-financial institutions (ECB_{NFI}). We then apply the ECB’s estimated share of portfolio at risks (ECB_{FR}) to derive the amount of German assets that are vulnerable to flood events (TA^F):

$$TA^F = TA \cdot ECB_{NFI} \cdot ECB_{FR} \quad (1)$$

Specifically, TA is the total amount of assets for our sample, ECB_{NFI} is equal to 59.8% and ECB_{FR} is set at 0.6%. Next, we allocate the German assets that are susceptible to flood events across different n NUTS2 regions, by taking into account the corresponding share of population exposed to river floods (P_n^F). This allows for a more accurate distribution of assets based on the level of vulnerability to flood events in each specific region. Essentially, regions with a higher proportion of the population exposed to these events will have a greater amount of assets at risk. While more detailed data at the NUTS3 level would provide more

⁶ See Table A.1 in Appendix A for descriptive statistics on the representativeness of the sample by country.

⁷ See Koetter et al. (2020) and Behr and Schmidt (2015).

⁸ RWA density = RWA over TA; Solvency = Regulatory K over RWA; Leverage = Regulatory K over TA.

⁹ Indicators NEAR (Normalised exposure at risk) and PEAR (potential exposure at risk) and as published here: https://www.ecb.europa.eu/pub/pdf/other/ecb.climate_change_indicators202301~47c4bbbc92.en.pdf

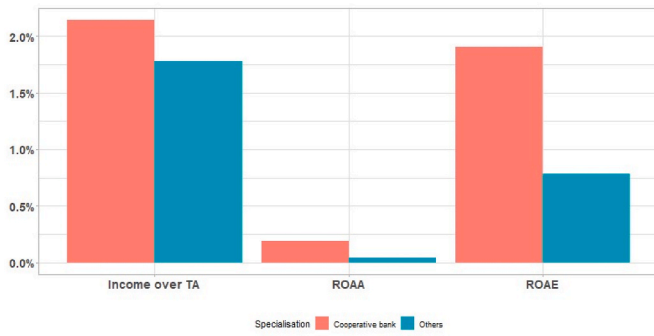


Fig. 3. Profitability indicators. The figure displays the comparison of selected profitability indicators - Net Income over total assets, return on average assets (ROAA) and return on average equity (ROAE) for a sample of 677 cooperative banks, and 47 commercial and 356 savings banks (“others”). The source of data is Orbis BankFocus, referring to the year 2022.

accurate estimates of the assets that may be at risk from acute physical factors, we are constrained by data limitations and thus maintain this assumption.¹⁰

The total assets of all banks in region n at risk of floods are represented by TA_n^F in the following equation:

$$TA_n^F = TA_n \cdot P_n^F \quad (2)$$

The share of population is obtained from the Risk Data Hub,¹¹ an EU-wide web-based geographical information system platform developed by the Joint Research Centre of the European Commission. The platform provides geo-referenced exposure data over one year¹² for various assets (such as buildings, population, critical services, and the environment).¹³ Fig. 4 presents the map for Germany along with a relative score measuring the population exposed in case of natural catastrophes. The figure shows that the exposure to river floods in Germany varies across different regions. Some regions, such as the areas along major rivers, are more susceptible to severe flooding than others. This is the case for Trier, Darmstadt, Niederbayern, Schwaben where population exposed to floods exceeds than 0.6%.

To account for the strain on banks' loan portfolios that can arise from the impact of river flood events, we consider the assets at risk of river flood-related risks as non-performing loans. This is also one assumption of the model, since part of the losses from non-performing loans might be recovered due to collateral for instance. All in all, when borrowers are affected by catastrophic events, which cause losses and damages to their businesses, the ability to honour their debt obligations becomes compromised, leading to a potential rise in unlikely-to-pay and non-performing loans. The total amount is thus distributed across each bank, denoted as i , based on their respective share of assets. Hence, we calculate the amount of bank assets at risk of becoming non-performing (NPL_i^F) as follow:

$$NPL_i^F = TA_n^F \cdot \frac{TA_i}{TA_n} \quad (3)$$

¹⁰ It is worth noting that the business area of regional banks may be smaller than NUTS2 areas, but larger than NUTS3 areas.

¹¹ <https://drmkc.jrc.ec.europa.eu/risk-data-hub>.

¹² The exposure of each asset over one year is an average of the exposure under different return periods weighted using the probability of occurrence. The “return periods” are estimates of the interval of time between events. For example, a return time of 100 years indicates that the event will occur once in 100 years on average, therefore the probability a similar event could occur in the same interval of time is 1% (1/100).

¹³ Each hazard is covered with a specific grid resolution (100m for river floods), and an aggregation at the level of local administrative units is also available.

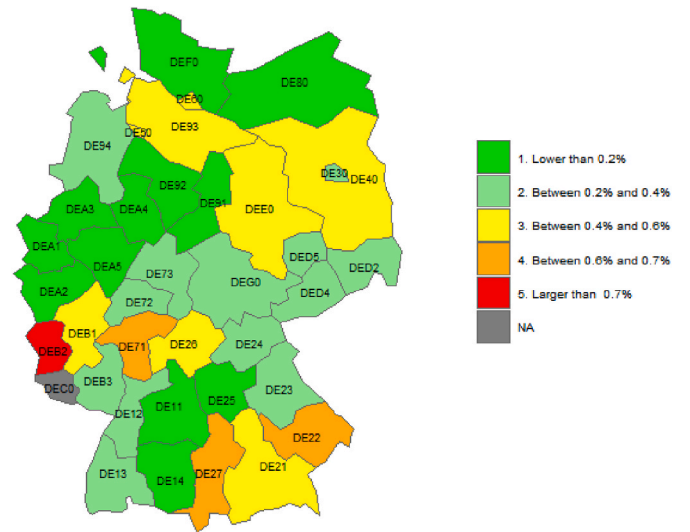


Fig. 4. Map of the population exposed to floods in Germany. Regions in red are more exposed to floods than regions in green. Source: own Calculation on JRC RDH data. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

where TA_i represents the assets of bank i , and TA_n represents the total assets of all banks in region n . Fig. 5 provides insights into the distribution of non-performing loans across banks. The graph reveals that the density distribution of NPL^F is skewed, indicating that the majority of institutions have a relatively size of NPL^F lower than 10% compared to their equity. This skewness suggests most banks in the sample might be able to manage credit risk and minimize the impact of loan defaults on their overall financial health.

As the global temperature raises, economic losses from flood-related events are expected to rise, which could potentially affect the worth of assets held by financial institution. For example, we can think about damage to assets serving as collateral, as in the case of mortgages given

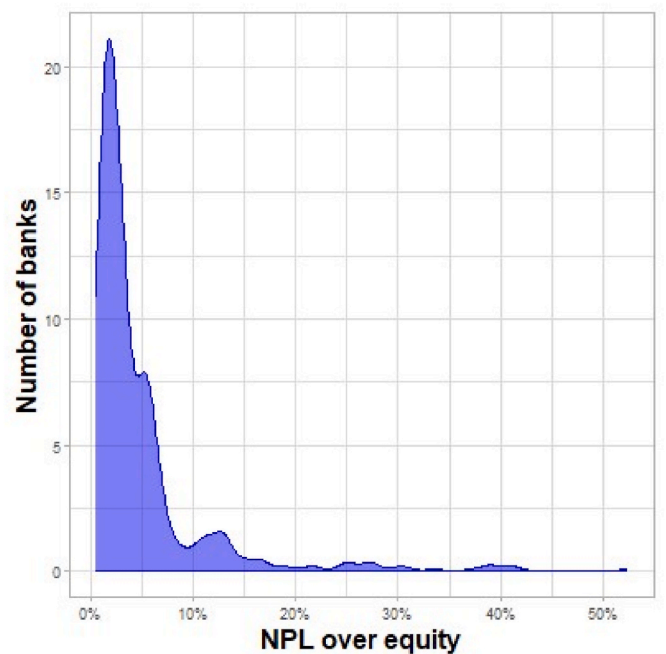


Fig. 5. Density distribution of NPL^F over equity across banks in the sample (under the actual climate condition). The source of data is Orbis BankFocus, referring to the year 2022, for a sample of 677 cooperative banks in Germany.

Table 1

Variation in economic losses under three temperature increases. Source: PESETA IV study (Feyen et al., 2022).

Climate Scenario	Increase
Increase under 1.5 °C	95%
Increase under 3 °C	304%
Increase with adaptation under 3 °C	80%

for properties located in vulnerable areas. Though accurately predicting the impact of global warming on the financial assets is challenging, it is reasonable to assume that as economic losses increase, assets currently exposed to flood risks, but not yet at risk, are likely to become non-performing in the future due to worsening climate conditions. To incorporate this assumption into our modelling framework, we assume that the size of river flood-related non-performing loan, across NUTS2 German regions, will increase proportionally to the severity of the economic impact caused by the rise in temperature:

$$NPL_{i,t}^F = NPL_i^F \cdot \Delta L_{t^C} \quad (4)$$

where:

- NPL_i^F are losses due to physical risk in the current situation;
- t^C refers to the increase in global temperature of a specific global warming scenario ($t = 1.5$ °C, 3.0 °C);
- ΔL_{t^C} indicates the increase in economic losses under a t^C increase in temperature with respect to today;
- NPL_{i,t^C}^F refers to the new flood-related losses under the warming scenario (1.5 °C, 3 °C).

Using country data from the PESETA IV study (Feyen et al., 2022), we determine the expected increase in overall economic losses resulting from flood events under a 1.5 °C and 3 °C increase in temperature, with and without adaptation strategies. Based on this information, we project the size of river floods-related non-performing loans under four global warming scenarios: current climate conditions, 1.5 °C increase without adaptation, 3 °C increase without adaptation, and 3 °C increase with adaptation strategies. These values are inputs to the model described in the next sections. Specifically, PESETA reveals that economic losses increase by 95% under a 1.5 °C increase in temperature, increase more than 300% under a 3 °C and adaptation strategies might reduce economic losses by 80% (See Table 1).

This assumption is a simplification that should be approached with caution. While it helps to address the complexity of the issue, it has the potential to oversimplify the situation, introducing uncertainty and potentially overlooking important factors.¹⁴ Specifically, we are assuming that the overall economy remains static, the structure of the banking system remains unchanged, and the risk appetite of financial institutions remains the same. It is important to recognize that in reality, these factors are subject to change and can significantly impact the relationship between economic losses and non-performing loans.

3. Simulation model

The modelling framework relies on the Systemic Model of Banking Originated Losses (SYMBOL, see De Lisa et al., 2011), a micro simulation portfolio model which uses bank-level data to simulate banking crisis scenarios. Specifically, individual banks may default based on their

actual capital (K) and on the probability of default attached to their portfolio (PD). The bank's portfolio PD is calculated by inverting the Basel formula, which relates the minimum capital requirements to the PD of a bank's portfolio and its RWA as follow:

$$CR(PD) = LGD \cdot N \left[\frac{\sqrt{RN^{-1} \left(\frac{0.08 \cdot RWA}{TA} \right)} + N^{-1}(PD)}{\sqrt{1-R}} \right] - PD, \quad (5)$$

where R_i is the correlation among exposures in the portfolio, LGD is the loss given default equal to 0.45 as per regulation,¹⁵ N is the normal distribution function.

Bank-specific PD s are then used to generate losses for individual banks via a Monte Carlo simulation, where randomness is introduced by sampling the underlying shocks. While provisions and write-offs should cover expected losses on an ongoing basis, unexpected losses refer to seldom but potentially significant losses that should be covered by capital. The failure of a bank is determined by comparing the size of these unexpected losses to the actual regulatory capital available for absorbing them. In other words, banks fail when simulated unexpected losses exceed the total actual capital.¹⁶ Some of the model iterations, particularly those where larger sampled shock, will likely result in at least one bank defaulting.

The output of the Monte Carlo simulation is an $I \times J$ matrix of unexpected losses GL_{ij} , obtained as follows:

$$GL_{ij} = LGD \cdot N \left[\sqrt{\frac{1}{1-R_i}} N^{-1}(PD_i) + \sqrt{\frac{R_i}{1-R_i}} N^{-1}(\alpha_{ij}) \right] - EL_i, \quad (6)$$

where the first term generates the total amount of losses and the second one (EL_i) approximates the expected losses.¹⁷ Specifically, $i = 1, \dots, I$ refers to the banks in the sample, $j = 1, \dots, J = 10,000$ denotes the model iteration. $N^{-1}(\alpha_{ij})$ are correlated normal random shocks. The α_{ij} shocks are correlated as they are defined as the sum of a common shock Z_j and a bank-specific shock W_{ij} , as follows:

$$N^{-1}(\alpha_{ij}) = l \cdot Z_j + \sqrt{1-l^2} \cdot W_{ij}, \quad (7)$$

where l are the loadings, W_{ij} are the idiosyncratic shocks, and Z_j is a common shock which might be linked, for example, with overall economic developments. The standard version of the model, which is used in this paper, sets l so as to yield a fixed correlation of 0.5 across the α_{ij} .¹⁸ The shocks Z_j and W_{ij} are drawn from a standardized normal distribution as in the Vasicek model (see Vasicek (1977); De Lisa et al. (2011)).

In each iteration, the following three components are summed up for each bank: i) losses that cannot be absorbed by capital, and ii) recapitalization needed to bring the banks back to a viability status, i.e. a total regulatory capital ratio of 8% of RWA s and iii) climate-related financial losses (L^F) to assess the impact of climate change on the bank's balance sheet. The framework subsequently assesses whether the banks' capital is high enough to absorb losses and climate-related losses or if these could potentially lead to additional failures within the banking sector.

From here on, we then define as losses for each bank and each iteration, the unexpected losses in excess of capital plus recapitalization needs ($ExLR_{ij}$), as follows:

$$ExLR_{ij} = \max(GL_{ij} - K_i + 8\%RWA_i + L_i^F, 0), \quad (8)$$

of course, L_i^F are taken into consideration just when physical risk materializes. Excess losses plus recapitalization needs for individual banks

¹⁴ See also the discussion about uncertainty in the Introduction. It is also worth noting that our assumptions simplify the complex interactions between climate change and bank finances. Additional factors, such as non-linearities and feedback effects, might lead to both an increase and a decrease in our estimates. Including additional factors in our analysis is particularly challenging due to data limitations. We thank an anonymous referee for these suggestions.

¹⁵ <https://www.fdic.gov/analysis/cfr/working-papers/2006/2006-10.pdf>

¹⁶ See Benczur et al. (2017) for details.

¹⁷ Expected loss is calculated as the product of PD , LGD , and total assets.

¹⁸ See Di Girolamo et al. (2017) for a discussion on a more sophisticated correlation structure.

yield the following aggregate loss distribution for each iteration:

$$L_j = \sum_{i=1}^I \text{ExLR}_{ij}. \quad (9)$$

Each point in the distribution (i.e. each iteration) is associated with a different level of financial/economic distress. Using individual banks' distribution, they can be aggregated at different scales (e.g. country, NUTS2, etc.).¹⁹ Final loss (FL) are then computed by taking the expected value of aggregate bank losses (as defined in Equation (9)) across selected iterations. This corresponds to the Expected Shortfall concept:

$$FL = \mathbb{E}_j[(L_j | j \text{ in } \hat{j})], \quad (10)$$

where the selection of iterations ($\hat{j}$) depends on whether we analyse a normal economic situation or a crisis type situation.

4. Physical risk as a trigger of a crisis

4.1. Modelling framework

In this application of the model, we investigate under which conditions river flood events could be the trigger of a crisis for cooperative banks in Germany even in the absence of a large economic shock. To do so, we start by comparing SYMBOL results with and without climate related financial losses to identify those banks that would fail due to a materialization of physical risk but would not fail otherwise. Essentially, based on the simulation results, we identify iterations where banks default when river-flood non-performing loans are taken into account but do not default without. Formally, these correspond to the runs satisfying the following condition:

$$\tilde{j} := j \text{ such that } \begin{cases} \text{ExLR}_{ij} = 0, \text{ without physical risk,} \\ \text{ExLR}_{ij} > 0, \text{ with physical risk.} \end{cases} \quad (11)$$

Final losses are calculated over these selected runs according to Equation (9), and climate-related losses (L^F in Equation (8)) account for both first and second-round effects.

The initial, or first-round, loss corresponds to the size of river-flood non-performing loans. These climate related losses could cause certain banks to default, and we assume that such defaults may initiate a sequence of devaluation dynamics for exposed assets. This is reflected on the one hand, in a lower share of exposed assets on banks' balance sheets, and on the other hand, in lower market prices for those assets. To capture this phenomena, we model a dynamic mechanism whereby, immediately upon recognition of the physical risks posed by river-flood events, a devaluation of exposed assets takes place, leading to further failures and second round losses in the banking sector. This devaluation cascade continues until the system reaches a new equilibrium, i.e. no more banks default. In mathematical terms, first and second-round climate-related losses (L^F in Equation (8)), are calculated as follows:

$$L_{i,t_r}^F = \begin{cases} NPL_i^F, & \text{if } t_r = 0, \\ NPL_i^F + \sum_{k=1}^{t_r} Dev_{i,k} \cdot TA_{i,e,k}^F, & \text{if } t_r > 0, \end{cases} \quad (12)$$

where i denotes the banks, t_r denotes a particular round within the devaluation cascade, $TA_{i,e}^F$ refers to the total value of assets exposed to river flood, and $Dev_{i,k}$ corresponds to the size of asset devaluation in each round. Notably:

$$TA_{i,e,t_r}^F = \begin{cases} 29.8\% TA, & \text{if } t_r = 0 \text{ as estimated by ECB,} \\ TA_{i,e,t_r-1}^F - L_{i,t_r-1}^F, & \text{if } t_r > 0. \end{cases} \quad (13)$$

The first devaluation of assets is proportional to the size of river floods-related non-performing loans in relation to the total value of exposed assets. In subsequent steps, the size of the devaluation is adjusted to reflect the variation observed in losses associated with climate-related events at the previous rounds of the devaluation cascade. In formula, the devaluation (Dev_k) is equal to:

$$Dev_{i,t_r} = \begin{cases} \frac{NPL_i^F}{TA_{i,e,0}^F}, & \text{if } t_r = 1, \\ \min\left(\frac{FL_{i,t_r-1} - FL_{i,t_r-2}}{TA_{i,e,t_r-1}^F}, 1\right), & \text{if } t_r > 1. \end{cases} \quad (14)$$

See Appendix B for the cascade routine flowchart.

4.2. Results

Based on the SYMBOL simulation and considering climate-related losses from river-flood events, our findings suggest that the impact on German cooperative banks is negligible today. However, this could become a more significant concern if the temperature increases by 1.5 °C and even worse under a 3 °C increase.

Table 2 presents the percentage of banks that default at each step, while Table 3 shows the corresponding percentage of assets affected. At step 0, our results indicate that physical risk initially leads to a relatively small number of failures in the banking system. Specifically, 0.2% of the total number of banks in our sample are in default, which also corresponds to 0.2% of total German assets. Under the most severe climate conditions, the initial share of failed banks increases to 7% of the total number of banks, affecting 5% of total assets (see Tables 2 and 3). The dynamic feature introduced to simulate the devaluation of exposed assets leads to more banks defaulting in the system. While the final number of defaults does not increase significantly under the current temperature level, the worst global warming scenario could result in the failure of 29% of banks, which hold approximately 24% of the total assets.

In terms of financial losses, the impact is relatively small under the

Table 2

Number of banks defaulted over the total number of banks in the sample (crisis triggered by physical risk).

Devaluation step	Current	1.5 °C increase	3 °C increase
Step 0	0.2%	3%	7%
Step 1	3%	7%	17%
Step 2	3%	7%	23%
Step 3	3%	8%	25%
Step 4	3%	8%	27%
Step 5	3%	8%	27%
Step 6	3%	8%	28%
Step 7	3%	8%	28%
Step 8	3%	8%	29%

¹⁹ This modelling framework can accommodate different degrees of comonomality by allowing for various shock correlation structures, while a contagion mechanism is not explicitly considered. We do not explicitly model contagion effects through the interbank market for three main reasons. First, contagion would introduce an additional layer of complexity, increasing the uncertainty around the results. Second, modelling contagion dynamics is quite challenging, as public data do not fit for the purpose of calibrating the network. Third, the comprehensive crisis management and deposit insurance framework put forward by the Commission after the last financial crisis (notably, the bail-in tools and the establishment of resolutions funds), is expected to prevent the spreading of contagion as distressed banks would be resolved, liquidated or recapitalised well before direct contagion effects could materialize (see, e.g. Benczur et al. (2017)).

Table 3

Assets of banks defaulted over the total amount of assets in the sample (crisis triggered by physical risk).

Devaluation step	Current	1.5 °C increase	3 °C increase
Step 0	0.2%	1%	5%
Step 1	1%	4%	15%
Step 2	1%	5%	19%
Step 3	1%	5%	22%
Step 4	1%	5%	23%
Step 5	1%	5%	24%
Step 6	1%	5%	24%
Step 7	1%	5%	24%
Step 8	1%	5%	24%

Table 4

Size of losses in the sample as share of total assets (crisis triggered by physical risk).

Devaluation step	Current	1.5 °C increase	3 °C increase
Step 0	0%	0%	0.1%
Step 1	0.02%	0.2%	0.6%
Step 2	0.02%	0.2%	0.9%
Step 3	0.02%	0.2%	1%
Step 4	0.02%	0.2%	1.1%
Step 5	0.02%	0.2%	1.1%
Step 6	0.02%	0.2%	1.1%
Step 7	0.02%	0.2%	1.1%
Step 8	0.02%	0.2%	1.1%

current climate conditions but could become significant as temperatures increase up to 3 °C. Specifically, aggregated losses amount to 0.02% of total assets at the present temperature level, 0.2% under a temperature increase of 1.5 °C, and 1.1% under a 3 °C increase (see Table 4).²⁰ These results show that should such risk materializes nowadays the recourse to public finances would be contained. However, an increase in temperature would pose challenges for the corporate sector that could destabilize the German regional banks, at least in some regions.

When examining specific NUTS2 regions, our results indicate that a few regions could be particularly strongly affected, with losses reaching up to 10% of total assets under a 1.5 °C temperature increase. Fig. 6 shows that there are at least three regions with a share of losses over total assets larger than 5%, even under a 1.5 °C temperature increase (and even higher under a 3 °C temperature increase). Comparing the two panels in Fig. 6, one can see that there are just few NUTS2 regions contributing to the overall losses.

5. A crisis not triggered by physical risk

5.1. Modelling framework

In this section, we use the model presented above to assess the impact of river-flood events, during a systemic banking crisis which is *not* triggered by climate crises. We do so by looking at the effect of including first-order climate-related losses (L^F) in Equation (8), without the amplification mechanism which is considered unnecessary in circumstances where financial stability is already compromised. Hence, L^F is equal to river floods-related non-performing loans (NPL^F).

We focus on the (very) right tail of the loss distribution, which is associated with a severe, but plausible, banking crisis. Technically, this

part of the distribution corresponds to values of Z_j which are farther than 3 standard deviations from the mean.²¹ This corresponds to the following iterations j :

$$\tilde{j} = \{j \text{ such that } Z_j > \mathbb{E}(Z_j) + 3\text{std}(Z_j)\}. \quad (15)$$

Final losses (FL) are then computed as in Equation (10) by taking the expected value of aggregate bank losses (as defined in Equation (9)) across the selected iterations. By comparing results with and without physical risk one can derive the size of final losses when river flood events materialize.

5.2. Results

In a crisis situation, approximately 14% of banks (representing nearly 15% of total assets) default, resulting in losses of 0.3% of total assets in the German banking system. The inclusion of climate-related factors exacerbates the situation, as physical risk contributes to more losses and failures. While the contribution of additional physical risks to overall losses is relatively modest under the current climate conditions, it becomes significant under a climate warming scenario. In terms of number of banks involved, Table 5 shows that flood events would cause a 4% increase in defaults among cooperative banks under the current climate situation, and a 17% increase under a temperature rise of up to 3 °C. In terms of losses, for 1.5 °C, aggregate losses, as a share of total assets, increase from 0.3% to 0.5% on average and up to 0.9% under a 3 °C (see Table 6).

Fig. 7 shows the increase in losses: the situation is quite heterogeneous across regions, with some NUTS2 subject to very mild (or almost zero) impacts and others where losses can increase substantially with respect to a situation with no climate risk. This is the case in particular for three regions, having a more than 5% increase for the 1.5 °C temperature increase scenario, and up to 30% in the 3 °C temperature increase scenario.

6. Adaptation strategies and impact on the banking sector

PESETA IV study (Feyen et al., 2022) quantifies that, under a 3 °C global temperature increase by the end of the century, in the EU + UK, river floods related losses would increase by six times, with more than half a million people exposed every year.²² The implementation of effective adaptation strategies plays a crucial role in mitigating economic losses and enhancing the resilience of firms to climate risks. By proactively implementing and integrating adaptation measures, businesses can minimize their vulnerability to climate-related events, thereby reducing the size of potential economic losses and enabling businesses to better anticipate or recover from climate-related shocks. The European Commission adopted its new EU Adaptation strategy on February 24, 2021. The four main principles of the strategy are a smarter adaptation, a systemic and faster set of actions, and international support for climate change adaptation and common policies.

Due to the uncertainty surrounding future scenarios, quantifying climate damages and determining effective adaptation strategies remains a difficult task. However, PESETA IV has taken an initial step by explicitly modeling adaptation strategies and estimating the potential reduction in losses. According to the report, adaptation measures would be very effective in reducing population exposed and potential economic losses, protecting citizens, preserving economic activities, and building a more resilient financial system.

The most effective adaptation strategy appears to be associated to

²⁰ It is important to note that in reality, the impact would be much more contained due to bank recovery mechanisms and troubled asset purchase programs that would be triggered before a large number of banks become insolvent.

²¹ As Z_j may be seen as a negative economic shock, this calibration is representative of a recession comparable to those observed on the occasion of the global financial crisis and the Covid-19 pandemic.

²² If one compares these numbers with the present situation, losses move from 7.8 to 48 billion per year and people exposed go from 172,000 to 4,820,000.

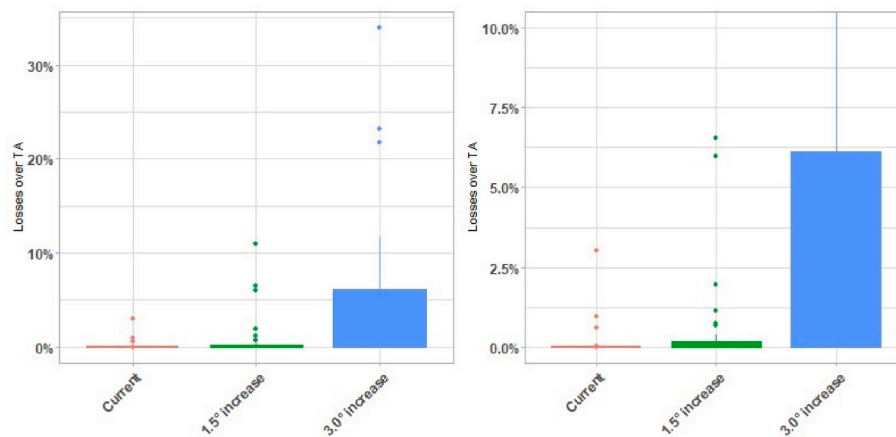


Fig. 6. Losses over TA distributions across NUTS2 regions under different temperature levels (crisis triggered by physical risk). Left panel: full sample, right panel: with no outliers. Note: The middle line is the median, the observation separating the sample in two halves. The extremes of the box are the first and third quartile, half of the sample is included inside the box, and a quarter is outside on each side. The whiskers are the highest or lowest observation lying within 1.5 times the length of the box from the median.

Table 5

Number of banks defaulted over the total number of banks in the sample and assets involved over the total amount of assets in the sample (systemic banking crisis).

	Without physical risk		With physical risk		
	Current		Current	1.5 °C increase	3 °C Increase
Number of banks	14%		18%	22%	31%
Assets	15%		18%	22%	30%

Table 6

Size of losses in the sample as share of TA (systemic banking crisis).

	Without physical risk	With physical risk		
	Current	Current	1.5 °C increase	3 °C Increase
Losses (% TA)	0.3%	0.4%	0.5%	0.9%

creating areas within or aside the river network that can be flooded in a controlled manner when the river stage reaches critical (see Arrighi et al. (2018), Dottori et al. (2020), and Appendix C for details). The reduction of flood peaks installing retention reservoirs could reduce by around EUR 40 billion the losses related to river floods, and around 400,000 fewer people would be exposed each year. Other adaptation measures such as the restoration of floodplain areas, strengthening the existing dyke system, and implementing building-based damage reduction measures are effective in further reducing the original impact. Overall, PESETA estimates point to a 80% reduction of the impact of climate-related events under a 3 °C (see Table 1).²³

In our modelling framework, we apply this reduction to the estimated amount of river flood-related non-performing loans under a 3 °C increase in temperature and we run our analysis to estimate the size of final losses under a normal economic situation and a banking crisis. Results show that the financial sector would benefit significantly, as the likelihood of either sudden asset devaluations and final losses driven by climate-related events diminishes to a point where they are comparable to the present circumstances, during a system crisis and in normal times

(see Figs. 8 and 9).

7. Conclusions

In this paper, we propose a methodology to assess the effect of river-flood events occurring within Germany on *regional banks*. Our findings show that the actual overall impact of physical risk seems to be more pronounced when it occurs during a pre-existing financial crisis. This holds true even without a dynamic mechanism that triggers the devaluation of assets located in area prone to flood events. Such a mechanism is considered unnecessary in circumstances where financial stability is already compromised, and the financial system is very susceptible to additional shocks. Notably, under a normal economic situation, only a few banks may default because of climate risk. However, we quantify that climate-related risks would increase overall losses by 0.1% under the current temperature level when considering a static systemic financial crisis scenario.

Should temperature rise, an increased perception of risk would activate a devaluation mechanism that would become disruptive when physical risk is the only factor causing instability. In such a situation, final banking losses become comparable to those during a systemic financial crisis. In fact, an increase in temperature is expected to cause more frequent and severe river floods, which could contribute to higher economic losses than currently experienced. For cooperative banks, this situation cause a surge in non performing loans as borrowers struggle to repay their loans. As temperatures rise, assets that are currently considered safe may become vulnerable to flood risk. In response, the market could devalue assets located in flood-prone areas, driven by an increase in risk perception. This devaluation would, in turn, amplify final losses for banks holding these assets, thus exacerbating the overall impact on the banking sector and making it similar to a systemic crisis. Notably, losses would raise up to 0.9% with a 3 °C temperature increase under a pre-existing crisis, with banks in some regions suffering much

²³ As a caveat, PESETA IV highlight that estimates are surrounded by relevant uncertainty and as such costs and benefit might be larger both in terms of economic losses and reductions due to adaptation strategies.

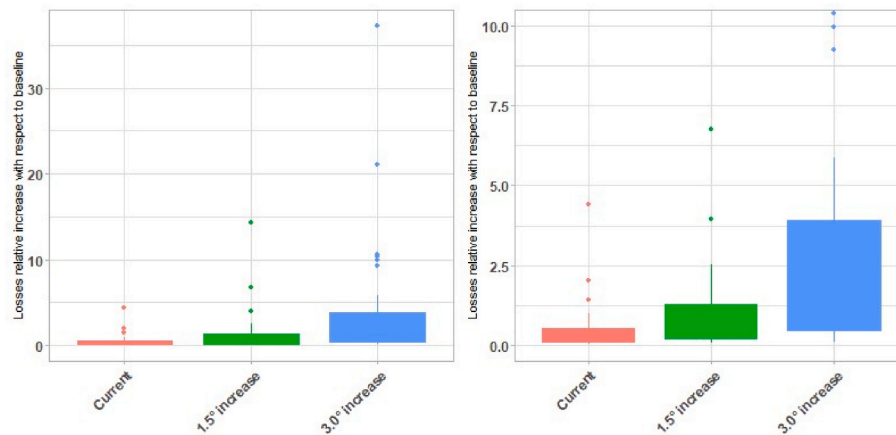


Fig. 7. Losses' relative increase with respect to Baseline distributions across NUTS2 regions, under three temperature levels (systemic banking crisis). Left panel: full sample, right panel: with no outliers.

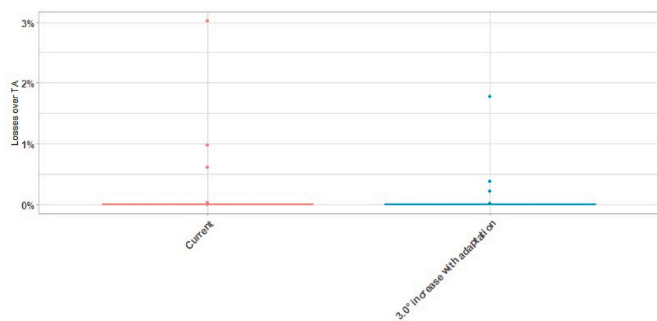


Fig. 8. Losses' distribution (crisis triggered by physical risk) for Current and 3.0 °C increase with adaptation.

higher losses than in others. Should a physical risk materialize in normal times, disorderly market adjustments would have a negligible impact under the actual warming conditions, however, could lead to significant

losses for the German regional banking system up to 1% of total assets if a 3 °C temperature rise materialized. In both cases, the variability of losses across NUTS2 regions, is quite large, with some regions reaching 5% of total assets in case of an existing financial crisis, and even to 10% in a normal economic situation. This negative impact can be effectively tackled with the implementation of adaptation strategies. Therefore our findings support the idea that banks operating in regions prone to such geographic risks should take measures to safeguard themselves. One potential approach is to implement additional bankspecific capital add-ons, which would enhance the resilience of the banking sector against these naturalrelated shocks. This strategy would help mitigate the potential negative impacts on banks' financial stability and ensure their ability to withstand and recover from such events. Alternatively, banks can contribute to the adaptation efforts aimed at reducing the vulnerability of the regions they operate in. By supporting adaptation initiatives, such as investing in climate-resilient infrastructure or providing financial services to support sustainable practices, banks can play a crucial role in building the resilience of the financial sector and reducing the overall impact on the economy.

Further research could investigate more in detail the overall impact of physical risk on the financial sector, as increasing attention is devoted both by policymakers and academics to the role of climate risk for

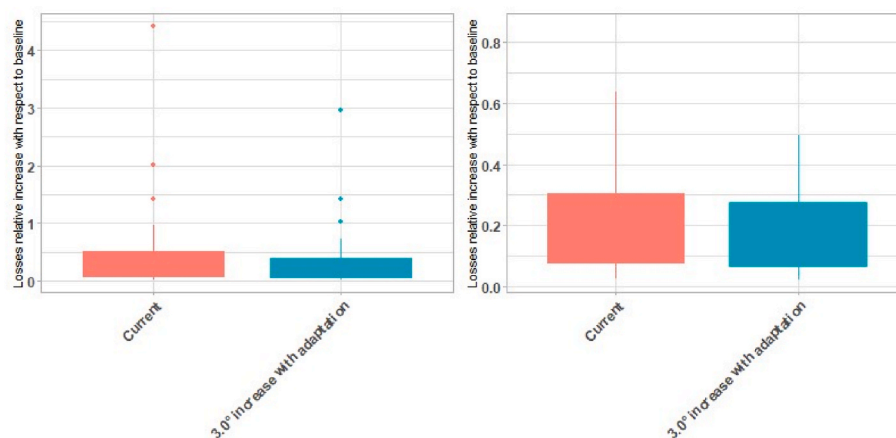


Fig. 9. Losses' (systemic crisis) relative increase with respect to Baseline distributions across NUTS2 regions for Current and 3.0 °C increase with adaptation. Left panel: full sample, right panel: with no outliers.

financial stability and debt sustainability.

CRedit authorship contribution statement

Mario Bellia: Writing – original draft, Formal analysis, Data curation, Conceptualization. **Erica Francesca Di Girolamo:** Writing – original draft, Formal analysis, Data curation, Conceptualization. **Andrea Pagano:** Writing – original draft, Formal analysis,

Conceptualization, Data curation.

Declaration of competing interest

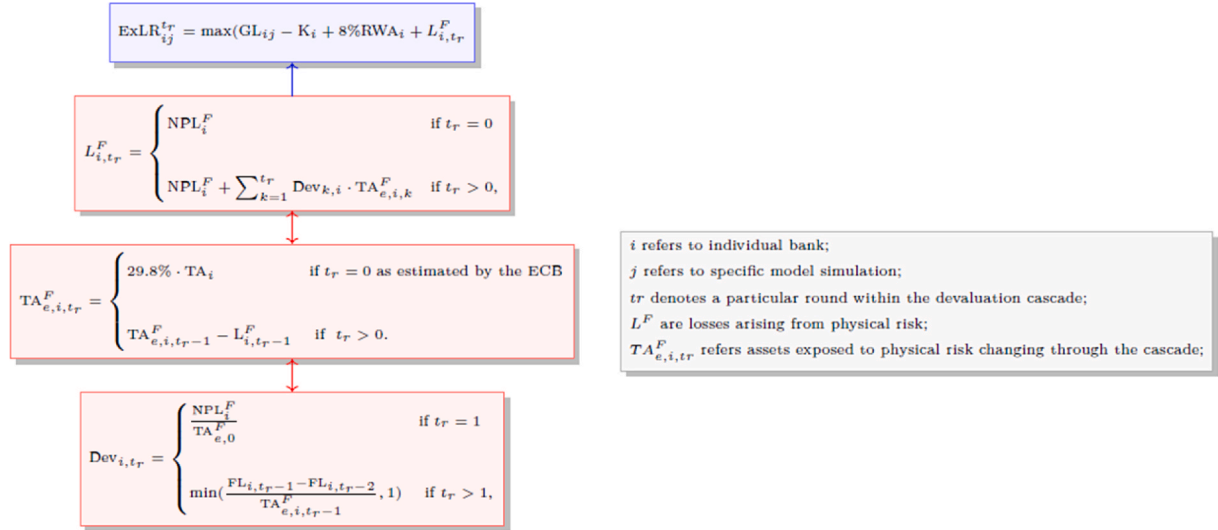
The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendices.

A Data inputs

Region	Banks	TA million €	FL _{NPL} million €	Region	Banks	TA million €	FL _{NPL} million €
DE11	49	72,369	96	DE91	10	15,011	31
DE12	34	66,291	153	DE92	11	24,954	59
DE13	21	37,213	71	DE93	14	10,609	130
DE14	30	21,936	41	DE94	45	34,474	106
DE21	51	70,189	344	DEA1	20	113,655	67
DE22	24	21,032	123	DEA2	21	33,839	97
DE23	26	29,112	52	DEA3	22	36	32
DE24	15	14,565	37	DEA4	16	25,426	24
DE25	21	26,778	15	DEA5	24	48,948	44
DE26	18	16,473	87	DEB1	19	19,191	132
DE27	39	35,207	179	DEB2	6	5,22	67
DE30	3	26,097	133	DEB3	16	39,714	67
DE40	11	6776	173	DED2	6	5445	74
DE50	2	1,91	52	DED4	6	6893	47
DE60	5	12,437	119	DED5	4	2678	42
DE71	19	33,79	417	DEE0	9	5111	190
DE72	12	16,673	45	DEF0	20	25,752	42
DE73	12	12,226	47	DEG0	10	7,6	87
DE80	6	4,82	19				

B Cascade routine's flow chart



C Impact of different adaptation measures in reducing climate-related economic losses

PESETA IV (see Feyen et al. (2022)) assesses how different adaptation measures could mitigate the effects of climate change, via a cost-benefit analysis of the four adaptation measures: (1) strengthening dykes, (2) retention areas, (3) damage reduction measures, and (4) relocation, and

under three different socioeconomic projections according to the ECFIN 2015 Ageing Report: (1) base economy, (2) economy as 2050, (3) economy as 2100.²⁴

Different adaptation measures and different socioeconomic assumptions, lead to different impact, as shown in the next two tables.

Table C.1
Expected annual population exposed

Economic scenario	EU27+UK	EU27
Base economy	171.800	163.000
Base economy 3 °C increase	520.000	487.700
Economy as 2050 3 °C increase	353.400	373.60
Economy as 2100 3 °C increase	481.800	444.000

Table C.2
Reduction in expected annual damage (EAD) as compared with the “no adaptation” scenario for the year 2100

Adaptation measure	EU27+UK
Strengthening dykes	68%
Retention	82%
Damage reduction measures	50%
Relocation	19%

Data availability

Data will be made available on request.

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²⁴ see The 2015 Ageing Report https://ec.europa.eu/economy_finance/publications/european_economy/2015/pdf/ee3_en.pdf.